Determinant, Eigenvalue and Eigenvector

Linear Algebra

Zhanxing Zhu

zhanxingzhu.github.io

December 6, 2024

Determinant

Nonsingular matrices

For any matrix, whether or not it is nonsingular is a key question. Recall that an $n \times n$ matrix T is nonsingular if and only if each of these holds:

- any system $T\vec{x} = \vec{b}$ has a solution and that solution is unique;
- Gauss-Jordan reduction of T yields an identity matrix;
- the rows of T form a linearly independent set;
- the columns of T form a linearly independent set, a basis for $\mathbb{R}^n$;
- any map that T represents is an isomorphism;
- an inverse matrix T^{-1} exists.

This part develops a formula to determine whether a matrix is nonsingular.

Determining nonsingularity is trivial for 1×1 matrices.

$$\begin{pmatrix} a \end{pmatrix} \quad \text{is nonsingular iff} \quad a \neq 0$$

Recall the matrix inverse formula for the 2×2 matrix.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{is nonsingular iff} \quad ad - bc \neq 0$$

We can produce the 3×3 formula as we did the prior one, although the computation is intricate (Try as an exercise.).

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \text{is nonsingular iff} \quad aei + bfg + cdh - hfa - idb - gec \neq 0$$

With these cases in mind, we posit a family of formulas: a, $ad - bc$, etc. For each n the formula defines a *determinant* function $\det_{n\times n} : \mathcal{M}_{n\times n} \to \mathbb{R}$ such that **an** $n\times n$ **matrix** T **is nonsingular if and only if** $\det_{n\times n}(T) \neq 0$.

The idea: defining a function by conditions

The prior slide gives a formula for the 1×1, 2×2, and 3×3 determinants. But while those three formulas are a help, they don't make clear what should be the general formula, for the $n\times n$ case.

So we will proceed by stating some conditions that a determinant function must satisfy. The conditions extrapolate from the 1×1, 2×2, and 3×3 cases. They will let us compute the determinant of a square matrix via Gauss's Method, which we know to be fast and easy.

Definition of determinant

Definition A $n\times n$ *determinant* is a function $\det\colon \mathcal{M}_{n\times n}\to\mathbb{R}$ such that

1) $\det(\vec{\rho}_1,\ldots,k\cdot\vec{\rho}_i+\vec{\rho}_j,\ldots,\vec{\rho}_n)=\det(\vec{\rho}_1,\ldots,\vec{\rho}_j,\ldots,\vec{\rho}_n)$ for $i\neq j$
2) $\det(\vec{\rho}_1,\ldots,\vec{\rho}_j,\ldots,\vec{\rho}_i,\ldots,\vec{\rho}_n)=-\det(\vec{\rho}_1,\ldots,\vec{\rho}_i,\ldots,\vec{\rho}_j,\ldots,\vec{\rho}_n)$ for $i\neq j$
3) $\det(\vec{\rho}_1,\ldots,k\vec{\rho}_i,\ldots,\vec{\rho}_n)=k\cdot\det(\vec{\rho}_1,\ldots,\vec{\rho}_i,\ldots,\vec{\rho}_n)$ for any scalar k
4) $\det(I)=1$ where I is an identity matrix

(the $\vec{\rho}$'s are the rows of the matrix). We often write $|T|$ for $\det(T)$.

Consequences of the definition

Key properties of determinant

1) A matrix with two identical rows has a determinant of zero.
2) A matrix with a zero row has a determinant of zero.
3) A matrix is nonsingular if and only if its determinant is nonzero.
4) The determinant of an echelon form matrix is the product down its diagonal.

Consequences of the definition

Key properties of deternimant

1) A matrix with two identical rows has a determinant of zero.
2) A matrix with a zero row has a determinant of zero.
3) A matrix is nonsingular if and only if its determinant is nonzero.
4) The determinant of an echelon form matrix is the product down its diagonal.

Verification

1). To verify the first sentence swap the two equal rows. The sign of the determinant changes but the matrix is the same and so its determinant is the same. Thus the determinant is zero.

Consequences of the definition

Key properties of deternimant

1) A matrix with two identical rows has a determinant of zero.
2) A matrix with a zero row has a determinant of zero.
3) A matrix is nonsingular if and only if its determinant is nonzero.
4) The determinant of an echelon form matrix is the product down its diagonal.

Verification

1). To verify the first sentence swap the two equal rows. The sign of the determinant changes but the matrix is the same and so its determinant is the same. Thus the determinant is zero.

2). For the second sentence multiply the zero row by two. That doubles the determinant but it also leaves the row unchanged, and hence leaves the determinant unchanged. Thus the determinant must be zero.

3). Do Gauss-Jordan reduction for the third sentence, $T \to \cdots \to \hat{T}$. By the first three properties the determinant of T is zero if and only if the determinant of $\hat{T}$ is zero (although the two could differ in sign or magnitude). A nonsingular matrix T Gauss-Jordan reduces to an identity matrix and so has a nonzero determinant. A singular T reduces to a $\hat{T}$ with a zero row; by the second property its determinant is zero.

3). Do Gauss-Jordan reduction for the third sentence, $T \to \cdots \to \hat{T}$. By the first three properties the determinant of T is zero if and only if the determinant of $\hat{T}$ is zero (although the two could differ in sign or magnitude). A nonsingular matrix T Gauss-Jordan reduces to an identity matrix and so has a nonzero determinant. A singular T reduces to a $\hat{T}$ with a zero row; by the second property its determinant is zero.

4). The fourth sentence has two cases. If the echelon form matrix is singular then it has a zero row. Thus it has a zero on its diagonal and the product down its diagonal is zero. By the third sentence of this result the determinant is zero and therefore this matrix's determinant equals the product down its diagonal.

If the echelon form matrix is nonsingular then none of its diagonal entries is zero. This means that we can divide by those entries and use condition (3) to get 1's on the diagonal.

$$\begin{vmatrix} t_{1,1} & t_{1,2} & & t_{1,n} \\ 0 & t_{2,2} & & t_{2,n} \\ & & \ddots & \\ 0 & & & t_{n,n} \end{vmatrix} = t_{1,1} \cdot t_{2,2} \cdots t_{n,n} \cdot \begin{vmatrix} 1 & t_{1,2}/t_{1,1} & & t_{1,n}/t_{1,1} \\ 0 & 1 & & t_{2,n}/t_{2,2} \\ & & \ddots & \\ 0 & & & 1 \end{vmatrix}$$

Then the Jordan half of Gauss-Jordan elimination leaves the identity matrix.

$$= t_{1,1} \cdot t_{2,2} \cdots t_{n,n} \cdot \begin{vmatrix} 1 & 0 & & 0 \\ 0 & 1 & & 0 \\ & & \ddots & \\ 0 & & & 1 \end{vmatrix} = t_{1,1} \cdot t_{2,2} \cdots t_{n,n} \cdot 1$$

So in this case also, the determinant is the product down the diagonal.

The conditions allow us to compute the determinant of a matrix using Gauss's Method.

Example On this matrix we can perform the Gauss's Method steps of $-2\rho_1+\rho_2$ and $-3\rho_1+\rho_3$, followed by $-(5/3)\rho_2+\rho_3$. Condition (1) says that these row combination operations leave the determinant unchanged.

$$\begin{vmatrix} 1 & 3 & -2 \\ 2 & 0 & 4 \\ 3 & -1 & 5 \end{vmatrix} = \begin{vmatrix} 1 & 3 & -2 \\ 0 & -6 & -8 \\ 0 & -10 & -11 \end{vmatrix} = \begin{vmatrix} 1 & 3 & -2 \\ 0 & -6 & -8 \\ 0 & 0 & -7/3 \end{vmatrix}$$

The conditions allow us to compute the determinant of a matrix using Gauss's Method.

Example On this matrix we can perform the Gauss's Method steps of $-2\rho_1 + \rho_2$ and $-3\rho_1 + \rho_3$, followed by $-(5/3)\rho_2 + \rho_3$. Condition (1) says that these row combination operations leave the determinant unchanged.

$$\begin{vmatrix} 1 & 3 & -2 \\ 2 & 0 & 4 \\ 3 & -1 & 5 \end{vmatrix} = \begin{vmatrix} 1 & 3 & -2 \\ 0 & -6 & -8 \\ 0 & -10 & -11 \end{vmatrix} = \begin{vmatrix} 1 & 3 & -2 \\ 0 & -6 & -8 \\ 0 & 0 & -7/3 \end{vmatrix}$$

The determinant is the product down the diagonal: $1 \cdot (-6) \cdot (-7/3) = 14$.

Example To see what happens with a row swap, consider this matrix. The swap changes the determinant's sign.

$$\begin{vmatrix} 0 & 3 & 1 \\ 1 & 2 & 0 \\ 1 & 5 & 2 \end{vmatrix} = -\begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 1 & 5 & 2 \end{vmatrix}$$

Finish by performing $-\rho_1 + \rho_3$ followed by $-\rho_2 + \rho_3$

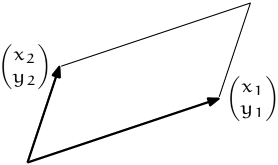
$$= -\begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 3 & 2 \end{vmatrix} = -\begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

and then multiplying down the diagonal. The determinant of the original matrix is -3.

Geometry of Determinants

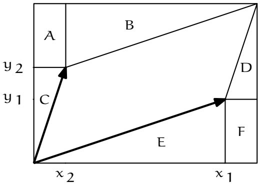
Box

This parallelogram is defined by the two vectors.



Definition In $\mathbb{R}^n$ the *box* (or *parallelepiped*) formed by $\langle \vec{v}_1, \ldots, \vec{v}_n \rangle$ is the set $\{ t_1\vec{v}_1 + \cdots + t_n\vec{v}_n \mid t_1, \ldots, t_n \in [0 \ldots 1] \}$.

Area



$$
\begin{aligned}
\text{box area} &= \text{rectangle area} - \text{area of } A - \cdots - \text{area of F} \\
&= (x_1 + x_2)(y_1 + y_2) - x_2 y_1 - x_1 y_1 / 2 \\
&\quad - x_2 y_2 / 2 - x_2 y_2 / 2 - x_1 y_1 / 2 - x_2 y_1 \\
&= x_1 y_2 - x_2 y_1
\end{aligned}
$$

The determinant of this matrix gives the size of the box formed by the matrix's columns.

$$
\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = x_1 y_2 - x_2 y_1
$$

Determinant as a function of the columns

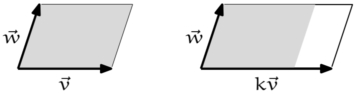
Because the determinant of the transpose equals the determinant of the matrix (e.g. checking the box area if swapping the coordinates of the two vectors), the row operation conditions in the definition translate over to column operation conditions. That is, (1) a determinant is unchanged by a column combination: where A is square and

$$A \xrightarrow{k\text{col}_i+\text{col}_j} \hat{A}$$

(with $i \neq j$) then $\det(A) = \det \hat{A}$, (2) a column swap changes the determinant's sign, and (3) multiplying a column by a scalar multiplies the entire determinant by that scalar. Condition (4), that the determinant of the identity matrix is 1, isn't about row or column operations so it still applies, without translation.

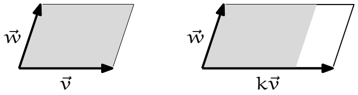
Geometric interpretation of the definition

We will argue that the conditions on a determinant function make good axioms for a function giving the size of a box. Consider the third condition, stated in column form, that rescaling a column rescales the entire determinant $\det(\ldots,k\vec{v}_i,\ldots)=k\cdot\det(\ldots,\vec{v}_i,\ldots)$. This fits because if we scale a column by a factor k then we expect that the box size also scales by that factor, as shown here.

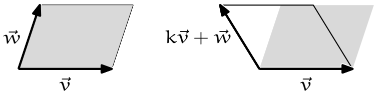


Geometric interpretation of the definition

We will argue that the conditions on a determinant function make good axioms for a function giving the size of a box. Consider the third condition, stated in column form, that rescaling a column rescales the entire determinant $\det(\ldots, k\vec{v}_i, \ldots) = k \cdot \det(\ldots, \vec{v}_i, \ldots)$. This fits because if we scale a column by a factor k then we expect that the box size also scales by that factor, as shown here.



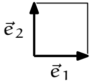
The determinant's first condition is that it is unaffected by combining columns. The picture



shows that the box formed by the two vectors $\vec{v}$ and $k\vec{v} + \vec{w}$ is slanted at a different angle than the one formed by $\vec{v}$ and $\vec{w}$ but the two boxes have the same base and height, and hence the same area.

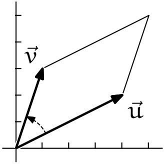
As we noted after the determinant's definition, its second condition is a consequence of the others so we leave it aside for a moment.

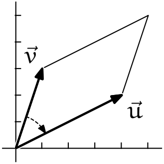
The final condition is that the determinant of the identity matrix is 1. This also fits with our program of arguing that the conditions in the definition of determinant are good ones for the function giving the size of the box formed by the columns of the matrix.



Orientation

Remark The second condition in the definition is that swapping changes the sign of the determinant. Consider these pictures, using the same pair of vectors.

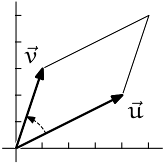




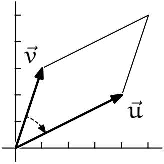
$$\begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix} = 10 \qquad \begin{vmatrix} 1 & 4 \\ 3 & 2 \end{vmatrix} = -10$$

Orientation

Remark The second condition in the definition is that swapping changes the sign of the determinant. Consider these pictures, using the same pair of vectors.



$$\begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix} = 10$$



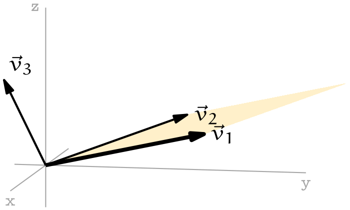
$$\begin{vmatrix} 1 & 4 \\ 3 & 2 \end{vmatrix} = -10$$

On the left, $\vec{u}$ is the first column in the matrix and we get a positive size. On the right, $\vec{v}$ is first and we get a negative size. On the left, the arc from first vector to second is counterclockwise while on the right, the arc from first to second is clockwise. The sign returned by the determinant reflects the *orientation* or *sense* of the box.

More on orientation: $\mathbb{R}^3$

These two vectors span a plane. The plane divides three-space into two parts, the part above and the part below.

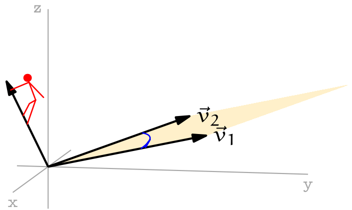
$$\vec{v}_1 = \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}$$



More on orientation: $\mathbb{R}^3$

These two vectors span a plane. The plane divides three-space into two parts, the part above and the part below.

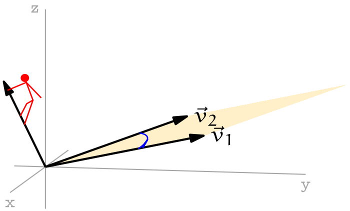
$$\vec{v}_1 = \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}$$



Take a $\vec{v}_3$ above that plane. More precisely, the $\vec{v}_3$ shown is on the side of the plane with the property that a person at the tip of $\vec{v}_3$ looking at the arc from $\vec{v}_1$ to $\vec{v}_2$ sees that arc as counterclockwise. Any such vector defines a box that is positive-sized.

$$\vec{v}_3 = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} \qquad \begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & -1 \\ 1 & 1 & 2 \end{vmatrix} = 25$$

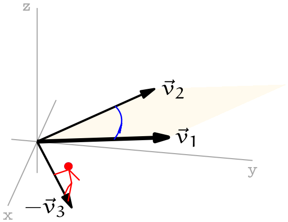
This is the *right hand rule*: place your right hand's pinky finger on the spanned plane so that your fingers curl from $\vec{v}_1$ to $\vec{v}_2$. Then vectors on the side of the plane with your thumb define positive-sized boxes.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

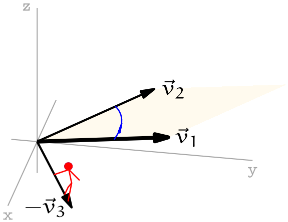
These pictures start above the plane, the perspective of the prior slide.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

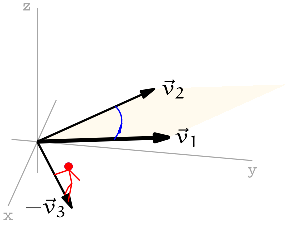
These pictures start above the plane, the perspective of the prior slide.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

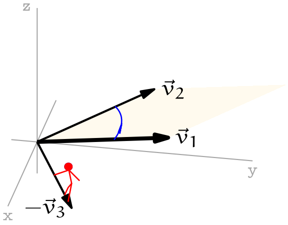
These pictures start above the plane, the perspective of the prior slide.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

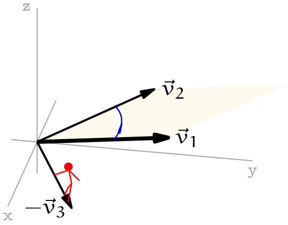
These pictures start above the plane, the perspective of the prior slide.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

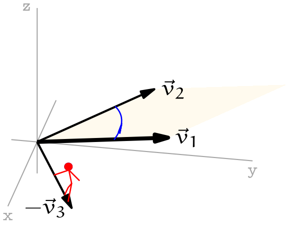
These pictures start above the plane, the perspective of the prior slide.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

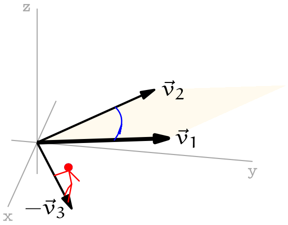
These pictures start above the plane, the perspective of the prior slide.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

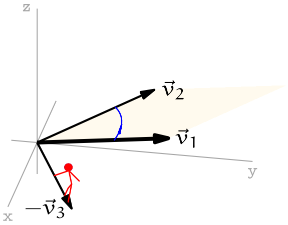
From below, to a person standing on $-\vec{v}_3$, the arc is clockwise.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

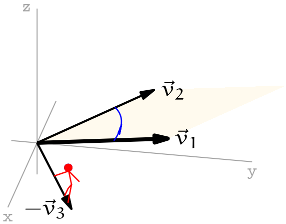
From below, to a person standing on $-\vec{v}_3$, the arc is clockwise.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

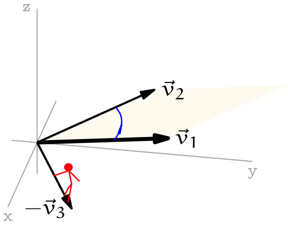
From below, to a person standing on $-\vec{v}_3$, the arc is clockwise.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

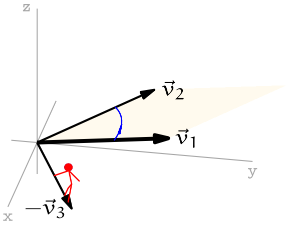
From below, to a person standing on $-\vec{v}_3$, the arc is clockwise.



A vector on the plane's other side, such as $-\vec{v}_3$, will have the same arc from $\vec{v}_1$ to $\vec{v}_2$ look clockwise and will define a negative-sized box.

$$\begin{vmatrix} 1 & -2 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & -2 \end{vmatrix} = -25$$

From below, to a person standing on $-\vec{v}_3$, the arc is clockwise.



To have it so that when you curl your fingers from $\vec{v}_1$ to $\vec{v}_2$ then the vector lies with your thumb, you must use your left hand.

Determinants are multiplicative

Theorem A transformation $t\colon \mathbb{R}^n \to \mathbb{R}^n$ changes the size of all boxes by the same factor, namely, the size of the image of a box $|t(S)|$ is $|T|$ times the size of the box $|S|$, where T is the matrix representing t with respect to the standard basis.

That is, the determinant of a product is the product of the determinants $|TS| = |T| \cdot |S|$.

Determinants are multiplicative

Theorem A transformation $t\colon \mathbb{R}^n \to \mathbb{R}^n$ changes the size of all boxes by the same factor, namely, the size of the image of a box $|t(S)|$ is $|T|$ times the size of the box $|S|$, where T is the matrix representing t with respect to the standard basis.

That is, the determinant of a product is the product of the determinants $|TS| = |T| \cdot |S|$.

Proof (**Optional**) First consider the case that T is singular and thus does not have an inverse. Observe that if TS is invertible then there is an M such that $(TS)M = I$, so $T(SM) = I$, and so T is invertible. The contrapositive of that observation is that if T is not invertible then neither is TS — if $|T| = 0$ then $|TS| = 0$.

Determinants are multiplicative

Theorem A transformation $t\colon \mathbb{R}^n \to \mathbb{R}^n$ changes the size of all boxes by the same factor, namely, the size of the image of a box $|t(S)|$ is $|T|$ times the size of the box $|S|$, where T is the matrix representing t with respect to the standard basis.

That is, the determinant of a product is the product of the determinants $|TS| = |T| \cdot |S|$.

Proof (**Optional**) First consider the case that T is singular and thus does not have an inverse. Observe that if TS is invertible then there is an M such that $(TS)M = I$, so $T(SM) = I$, and so T is invertible. The contrapositive of that observation is that if T is not invertible then neither is TS — if $|T| = 0$ then $|TS| = 0$.

Now consider the case that T is nonsingular. Any nonsingular matrix factors into a product of elementary matrices $T = E_1 E_2 \cdots E_r$. To finish this argument we will verify that $|ES| = |E| \cdot |S|$ for all matrices S and elementary matrices E. The result will then follow because $|TS| = |E_1 \cdots E_r S| = |E_1| \cdots |E_r| \cdot |S| = |E_1 \cdots E_r| \cdot |S| = |T| \cdot |S|$.

There are three types of elementary matrix. We will cover the $M_i(k)$ case; the $P_{i,j}$ and $C_{i,j}(k)$ checks are similar. The matrix $M_i(k)S$ equals S except that row i is multiplied by k. The third condition of determinant functions then gives that $|M_i(k)S| = k \cdot |S|$. But $|M_i(k)| = k$, again by the third condition because $M_i(k)$ is derived from the identity by multiplication of row i by k. Thus $|ES| = |E| \cdot |S|$ holds for $E = M_i(k)$. QED

Example The transformation $t_\theta\colon \mathbb{R}^2 \to \mathbb{R}^2$ that rotates all vectors through a counterclockwise angle θ is represented by this matrix.

$$T_\theta = \text{Rep}_{\mathcal{E}_2,\mathcal{E}_2}(t_\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & cos\theta \end{pmatrix}$$

Observe that t_θ doesn't change the size of any boxes, it just rotates the entire box as a rigid whole. Note that $|T_\theta| = 1$.

Example The transformation $t_\theta\colon \mathbb{R}^2 \to \mathbb{R}^2$ that rotates all vectors through a counterclockwise angle θ is represented by this matrix.

$$T_\theta = \mathrm{Rep}_{\mathcal{E}_2,\mathcal{E}_2}(t_\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

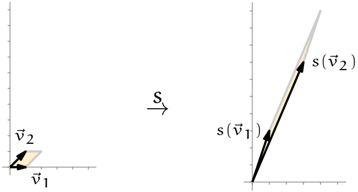
Observe that t_θ doesn't change the size of any boxes, it just rotates the entire box as a rigid whole. Note that $|T_\theta| = 1$.

Example The linear transformation $s\colon \mathbb{R}^2 \to \mathbb{R}^2$ represented with respect to the standard basis by this matrix

$$S = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

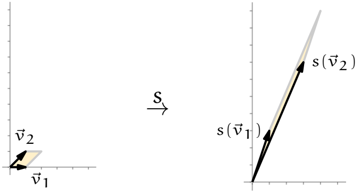
will, by the theorem, change the size of a box by a factor of $|S| = -2$. Here is s acting on a typical box.

The box defined by the two vectors $\vec{v}_1 = \binom{1}{0}$ and $\vec{v}_2 = \binom{1}{1}$ is transformed by s to the box defined by the two vectors $s(\vec{v}_1) = \binom{1}{3}$ and $s(\vec{v}_2) = \binom{3}{7}$.



Note the change in orientation, matching that the determinant is negative.

The box defined by the two vectors $\vec{v}_1 = \binom{1}{0}$ and $\vec{v}_2 = \binom{1}{1}$ is transformed by s to the box defined by the two vectors $s(\vec{v}_1) = \binom{1}{3}$ and $s(\vec{v}_2) = \binom{3}{7}$.



Note the change in orientation, matching that the determinant is negative.

The two sizes are easy.

$$\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1 \qquad \begin{vmatrix} 1 & 3 \\ 3 & 7 \end{vmatrix} = -2$$

Determinant of the inverse

Corollary If a matrix is invertible then the determinant of its inverse is the inverse of its determinant $|T^{-1}| = 1/|T|$.

Determinant of the inverse

Corollary If a matrix is invertible then the determinant of its inverse is the inverse of its determinant $|T^{-1}| = 1/|T|$.

Proof $1 = |I| = |TT^{-1}| = |T| \cdot |T^{-1}|$ QED

Example These matrices are inverse.

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2 \qquad \begin{vmatrix} -2 & 1 \\ 3/2 & -1/2 \end{vmatrix} = -1/2$$

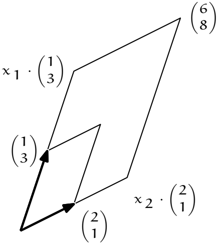
Cramer’s Rule

Geometric interpretation of linear systems

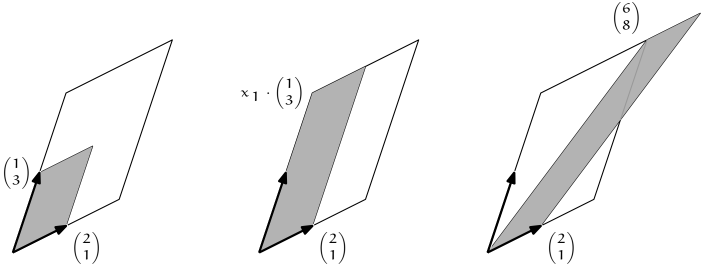
A linear system is equivalent to a linear relationship among vectors.

$$\begin{aligned} x_1 + 2x_2 &= 6 \\ 3x_1 + x_2 &= 8 \end{aligned} \qquad \Longleftrightarrow \qquad x_1 \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} + x_2 \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$$

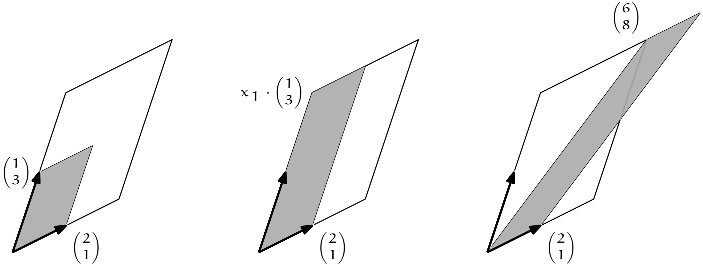
This drawing restates the algebraic question of finding the solution of a linear system into geometric terms: by what factors x_1 and x_2 must we dilate the sides of the starting parallelogram so that it will fill the other one?



Consider expanding only one side of the parallelogram. Compare the sizes of these shaded boxes.



Consider expanding only one side of the parallelogram. Compare the sizes of these shaded boxes.



Taken together we have this.

$$x_1 \cdot \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = \begin{vmatrix} x_1 \cdot 1 & 2 \\ x_1 \cdot 3 & 1 \end{vmatrix} = \begin{vmatrix} x_1 \cdot 1 + x_2 \cdot 2 & 2 \\ x_1 \cdot 3 + x_2 \cdot 1 & 1 \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 8 & 1 \end{vmatrix}$$

(The second last equality is by the first condition in the definition of determinants, applied to columns.)

Solving gives the value of one of the variables.

$$x_1 = \frac{\begin{vmatrix} 6 & 2 \\ 8 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}} = \frac{-10}{-5} = 2$$

Solving gives the value of one of the variables.

$$x_1 = \frac{\begin{vmatrix} 6 & 2 \\ 8 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}} = \frac{-10}{-5} = 2$$

The symmetric argument for the other side gives the other value.

$$x_2 = \frac{\begin{vmatrix} 1 & 6 \\ 3 & 8 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}} = 2$$

Cramer's Rule

Theorem Let A be an $n \times n$ matrix with a nonzero determinant, let $\vec{b}$ be an n-tall column vector, and consider the linear system $A\vec{x} = \vec{b}$. For any $i \in [1, \ldots, n]$ let B_i be the matrix obtained by substituting $\vec{b}$ for column i of A. Then the value of the i-th unknown is $x_i = |B_i|/|A|$.

Of course, if the matrix has a zero determinant then the system does not have a unique solution.

Example This system

$$\begin{aligned} 2x_1 + x_2 - x_3 &= 4 \\ x_1 + 3x_2 &= 2 \\ x_2 - 5x_3 &= 0 \end{aligned}$$

is

$$\begin{pmatrix} 2 & 1 & -1 \\ 1 & 3 & 0 \\ 0 & 1 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}$$

and

$$|A| = \begin{vmatrix} 2 & 1 & -1 \\ 1 & 3 & 0 \\ 0 & 1 & -5 \end{vmatrix} = -26 \qquad |B_2| = \begin{vmatrix} 2 & 4 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -5 \end{vmatrix} = 0$$

so $x_2 = 0/-26 = 0$.

A caution

Cramer's Rule is an interesting application of the geometry. And, it allows us to mentally solve small systems with simple numbers. But don't use it for systems having many variables; taking a determinant of a general large matrix is very slow.

Laplace’s formula for computing determinant

Minor

Definition For any $n \times n$ matrix T, the $(n-1) \times (n-1)$ matrix formed by deleting row i and column j of T is the i, j *minor* of T. The i, j *cofactor* $T_{i,j}$ of T is $(-1)^{i+j}$ times the determinant of the i, j minor of T.

$$\begin{pmatrix} t_{1,1} & t_{1,2} & t_{1,3} \\ t_{2,1} & t_{2,2} & t_{2,3} \\ t_{3,1} & t_{3,2} & t_{3,3} \end{pmatrix} \quad \begin{pmatrix} t_{1,1} & t_{1,2} & t_{1,3} \\ t_{2,1} & t_{2,2} & t_{2,3} \\ t_{3,1} & t_{3,2} & t_{3,3} \end{pmatrix} \quad \begin{pmatrix} t_{1,1} & t_{1,2} & t_{1,3} \\ t_{2,1} & t_{2,2} & t_{2,3} \\ t_{3,1} & t_{3,2} & t_{3,3} \end{pmatrix}$$

Minor

Definition For any $n \times n$ matrix T, the $(n-1) \times (n-1)$ matrix formed by deleting row i and column j of T is the i, j *minor* of T. The i, j *cofactor* $T_{i,j}$ of T is $(-1)^{i+j}$ times the determinant of the i, j minor of T.

$$\begin{pmatrix} t_{1,1} & t_{1,2} & t_{1,3} \\ t_{2,1} & t_{2,2} & t_{2,3} \\ t_{3,1} & t_{3,2} & t_{3,3} \end{pmatrix} \quad \begin{pmatrix} t_{1,1} & t_{1,2} & t_{1,3} \\ t_{2,1} & t_{2,2} & t_{2,3} \\ t_{3,1} & t_{3,2} & t_{3,3} \end{pmatrix} \quad \begin{pmatrix} t_{1,1} & t_{1,2} & t_{1,3} \\ t_{2,1} & t_{2,2} & t_{2,3} \\ t_{3,1} & t_{3,2} & t_{3,3} \end{pmatrix}$$

Example For this matrix

$$S = \begin{pmatrix} 3 & 1 & 2 \\ 5 & 4 & -1 \\ 7 & 0 & -3 \end{pmatrix}$$

the $2, 3$ minor is

$$\begin{pmatrix} 3 & 1 \\ 7 & 0 \end{pmatrix}$$

so the associated cofactor is $S_{2,3} = (-1)^5 \cdot (-7) = 7$.

Laplace's formula

Theorem Where T is an $n \times n$ matrix, we can find the determinant by expanding by cofactors on any row i or column j.

$$
\begin{aligned}
|T| &= t_{i,1} \cdot T_{i,1} + t_{i,2} \cdot T_{i,2} + \cdots + t_{i,n} \cdot T_{i,n} \\
&= t_{1,j} \cdot T_{1,j} + t_{2,j} \cdot T_{2,j} + \cdots + t_{n,j} \cdot T_{n,j} \\
&= \sum_{i=1}^{n} (-1)^{(i+j)} t_{ij} \det(M_{ij})
\end{aligned}
$$

Laplace's formula

Theorem Where T is an $n \times n$ matrix, we can find the determinant by expanding by cofactors on any row i or column j.

$$\begin{aligned}
|T| &= t_{i,1} \cdot T_{i,1} + t_{i,2} \cdot T_{i,2} + \cdots + t_{i,n} \cdot T_{i,n} \\
&= t_{1,j} \cdot T_{1,j} + t_{2,j} \cdot T_{2,j} + \cdots + t_{n,j} \cdot T_{n,j} \\
&= \sum_{i=1}^{n} (-1)^{(i+j)} t_{ij} \det(M_{ij})
\end{aligned}$$

We can find this determinant

$$\begin{vmatrix} 3 & 1 & 2 \\ 5 & 4 & -1 \\ 7 & 0 & -3 \end{vmatrix}$$

by expanding along the second row. Earlier we found $S_{2,3} = 7$ and the other two cofactors are here.

$$S_{2,1} = (-1)^3 \cdot \begin{vmatrix} 1 & 2 \\ 0 & -3 \end{vmatrix} = 3 \quad S_{2,2} = (-1)^4 \cdot \begin{vmatrix} 3 & 2 \\ 7 & -3 \end{vmatrix} = -23$$

The Laplace expansion gives $5 \cdot 3 + 4 \cdot (-23) - 1 \cdot 7 = -84$.

Matrix Similarity

We've defined two matrices H and $\hat{H}$ to be matrix equivalent if there are nonsingular P and Q such that $\hat{H} = PHQ$. We were motivated by this diagram showing H and $\hat{H}$ both representing a map h, but with respect to different pairs of bases, B, D and $\hat{B}, \hat{D}$.

$$
\begin{tikzcd}
V_{\mathit{wrt}\ B} \arrow[r, "h", "H"'] \arrow[d, "\text{id}"'] & W_{\mathit{wrt}\ D} \arrow[d, "\text{id}"'] \\
V_{\mathit{wrt}\ \hat{B}} \arrow[r, "h", "\hat{H}"'] & W_{\mathit{wrt}\ \hat{D}}
\end{tikzcd}
$$

We've defined two matrices H and $\hat{H}$ to be matrix equivalent if there are nonsingular P and Q such that $\hat{H} = PHQ$. We were motivated by this diagram showing H and $\hat{H}$ both representing a map h, but with respect to different pairs of bases, B, D and $\hat{B}, \hat{D}$.

$$\begin{tikzcd}
V_{wrt\ B} \arrow[r, "h", "H"'] \arrow[d, "\text{id}"'] & W_{wrt\ D} \arrow[d, "\text{id}"'] \\
V_{wrt\ \hat{B}} \arrow[r, "h", "\hat{H}"'] & W_{wrt\ \hat{D}}
\end{tikzcd}$$

We now consider the special case of transformations, where the codomain equals the domain, and we add the requirement that the codomain's basis equals the domain's basis. So, we are considering representations with respect to B, B and D, D.

$$\begin{tikzcd}
V_{wrt\ B} \arrow[r, "t", "T"'] \arrow[d, "\text{id}"'] & V_{wrt\ B} \arrow[d, "\text{id}"'] \\
V_{wrt\ D} \arrow[r, "t", "\hat{T}"'] & V_{wrt\ D}
\end{tikzcd}$$

In matrix terms, $\text{Rep}_{D,D}(t) = \text{Rep}_{B,D}(\text{id})\ \text{Rep}_{B,B}(t)\ \left(\text{Rep}_{B,D}(\text{id})\right)^{-1}$.

Similar matrices

Definition The matrices T and $\hat{T}$ are *similar* if there is a nonsingular P such that $\hat{T} = PTP^{-1}$.

Example Consider the derivative map $d/dx\colon \mathcal{P}_2 \to \mathcal{P}_2$. Fix the basis $B = \langle 1, x, x^2 \rangle$ and the basis $D = \langle 1, 1+x, 1+x+x^2 \rangle$. In this arrow diagram we will first get T, and then calculate $\hat{T}$ from it.

$$\begin{tikzcd}
V_{\text{wrt } B} \arrow[r, "t", "T"'] \arrow[d, "\mathrm{id}"'] & V_{\text{wrt } B} \arrow[d, "\mathrm{id}"'] \\
V_{\text{wrt } D} \arrow[r, "t", "\hat{T}"'] & V_{\text{wrt } D}
\end{tikzcd}$$

Similar matrices

Definition The matrices T and $\hat{T}$ are *similar* if there is a nonsingular P such that $\hat{T} = PTP^{-1}$.

Example Consider the derivative map $d/dx\colon \mathcal{P}_2 \to \mathcal{P}_2$. Fix the basis $B = \langle 1, x, x^2\rangle$ and the basis $D = \langle 1, 1+x, 1+x+x^2\rangle$. In this arrow diagram we will first get T, and then calculate $\hat{T}$ from it.

$$\begin{array}{ccc} V_{\textit{wrt}\ B} & \xrightarrow[T]{t} & V_{\textit{wrt}\ B} \\ {\scriptstyle \text{id}}\big\downarrow & & {\scriptstyle \text{id}}\big\downarrow \\ V_{\textit{wrt}\ D} & \xrightarrow[\hat{T}]{t} & V_{\textit{wrt}\ D} \end{array}$$

The action of d/dx on the elements of the basis B is $1 \mapsto 0$, $x \mapsto 1$, and $x^2 \mapsto 2x$.

$$\text{Rep}_B(\frac{d}{dx}(1)) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{Rep}_B(\frac{d}{dx}(x)) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{Rep}_B(\frac{d}{dx}(x^2)) = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$$

So we have this matrix representation of the map.

$$\mathsf{T} = \mathsf{Rep}_{B,B}\left(\frac{d}{dx}\right) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

So we have this matrix representation of the map.

$$T = \text{Rep}_{B,B}(\frac{d}{dx}) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

The matrix changing bases from B to D is $\text{Rep}_{B,D}(\text{id})$. We find these by eye

$$\text{Rep}_D(\text{id}(1)) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{Rep}_D(\text{id}(x)) = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \quad \text{Rep}_D(\text{id}(x^2)) = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

to get this.

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \qquad P^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Now, by following the arrow diagram we have $\hat{T} = PTP^{-1}$.

$$\hat{T} = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

To check that, and to underline what the arrow diagram says

$$\begin{tikzcd} V_{\mathit{wrt}\ B} \arrow[r, "t", "T"'] \arrow[d, "\mathrm{id}"'] & V_{\mathit{wrt}\ B} \arrow[d, "\mathrm{id}"'] \\ V_{\mathit{wrt}\ D} \arrow[r, "t", "\hat{T}"'] & V_{\mathit{wrt}\ D} \end{tikzcd}$$

we calculate $\hat{T}$ directly. The effect of the map on the basis elements is $d/dx(1) = 0$, $d/dx(1+x) = 1$, and $d/dx(1+x+x^2) = 1+2x$. Representing of those with respect to D

$$\text{Rep}_D(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{Rep}_D(1) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{Rep}_D(1+2x) = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}$$

gives the same matrix $\hat{T} = \text{Rep}_{D,D}(d/dx)$ as above.

The definition doesn't require that we consider the underlying maps. We can just multiply matrices.

Example Where

$$T = \begin{pmatrix} 0 & -1 & -2 \\ 2 & 3 & 2 \\ 4 & 5 & 2 \end{pmatrix} \qquad P = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

(note that P is nonsingular) we can compute this $\hat{T} = PTP^{-1}$.

$$\hat{T} = \begin{pmatrix} 2 & 0 & 0 \\ 3 & 1 & 4/3 \\ 27/2 & 3/2 & 2 \end{pmatrix}$$

The definition doesn't require that we consider the underlying maps. We can just multiply matrices.

Example Where

$$T = \begin{pmatrix} 0 & -1 & -2 \\ 2 & 3 & 2 \\ 4 & 5 & 2 \end{pmatrix} \qquad P = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

(note that P is nonsingular) we can compute this $\hat{T} = PTP^{-1}$.

$$\hat{T} = \begin{pmatrix} 2 & 0 & 0 \\ 3 & 1 & 4/3 \\ 27/2 & 3/2 & 2 \end{pmatrix}$$

Example The only matrix similar to the zero matrix is itself: $PZP^{-1} = PZ = Z$. The identity matrix has the same property: $PIP^{-1} = PP^{-1} = I$.

Similarity is an equivalence

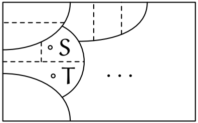
Similarity is an equivalence

Since matrix similarity is a special case of matrix equivalence, if two matrices are similar then they are matrix equivalent. What about the converse: must any two matrix equivalent square matrices be similar? No; the matrix equivalence class of an identity consists of all nonsingular matrices of that size while the prior example shows that an identity matrix is alone in its similarity class.

Similarity is an equivalence

Since matrix similarity is a special case of matrix equivalence, if two matrices are similar then they are matrix equivalent. What about the converse: must any two matrix equivalent square matrices be similar? No; the matrix equivalence class of an identity consists of all nonsingular matrices of that size while the prior example shows that an identity matrix is alone in its similarity class.

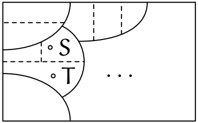
So some matrix equivalence classes split into two or more similarity classes — similarity gives a finer partition than does equivalence. This pictures some matrix equivalence classes subdivided into similarity classes.



Similarity is an equivalence

Since matrix similarity is a special case of matrix equivalence, if two matrices are similar then they are matrix equivalent. What about the converse: must any two matrix equivalent square matrices be similar? No; the matrix equivalence class of an identity consists of all nonsingular matrices of that size while the prior example shows that an identity matrix is alone in its similarity class.

So some matrix equivalence classes split into two or more similarity classes — similarity gives a finer partition than does equivalence. This pictures some matrix equivalence classes subdivided into similarity classes.



We naturally want a canonical form to represent the similarity classes. Some classes, but not all, are represented by a diagonal form.

Diagonalizability

Definition A transformation is *diagonalizable* if it has a diagonal representation with respect to the same basis for the codomain as for the domain. A *diagonalizable matrix* is one that is similar to a diagonal matrix: T is diagonalizable if there is a nonsingular P such that PTP^{-1} is diagonal.

Example This matrix

$$\begin{pmatrix} 6 & -1 & -1 \\ 2 & 11 & -1 \\ -6 & -5 & 7 \end{pmatrix}$$

is diagonalizable by using this

$$P = \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ -1/2 & 1/4 & 1/4 \\ -1/2 & -3/4 & 1/4 \end{pmatrix} \quad P^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \end{pmatrix}$$

to get this $D = PSP^{-1}$.

$$D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 12 \end{pmatrix}$$

Example This matrix is not diagonalizable.

$$N = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

The fact that N is not the zero matrix means that it cannot be similar to the zero matrix, because the zero matrix is similar only to itself. Thus if N were to be similar to a diagonal matrix D then D would have at least one nonzero entry on its diagonal.

The crucial point is that a power of N is the zero matrix, specifically N^2 is the zero matrix. This implies that for any map n represented by N with respect to some B, B, the composition $n \circ n$ is the zero map. This in turn implies that any matrix representing n with respect to some $\hat{B}, \hat{B}$ has a square that is the zero matrix. But for any nonzero diagonal matrix D^2, the entries of D^2 are the squares of the entries of D, so D^2 cannot be the zero matrix. Thus N is not diagonalizable.

Lemma A transformation t is diagonalizable if and only if there is a basis $B = \langle \vec{\beta}_1, \ldots, \vec{\beta}_n \rangle$ and scalars $\lambda_1, \ldots, \lambda_n$ such that $t(\vec{\beta}_i) = \lambda_i \vec{\beta}_i$ for each i. Then the corresponding matrix is also diagonalizable.

Lemma A transformation t is diagonalizable if and only if there is a basis $B = \langle \vec{\beta}_1, \dots, \vec{\beta}_n \rangle$ and scalars $\lambda_1, \dots, \lambda_n$ such that $t(\vec{\beta}_i) = \lambda_i \vec{\beta}_i$ for each i. Then the corresponding matrix is also diagonalizable.

Proof (**Optional**) Consider a diagonal representation matrix.

$$\mathrm{Rep}_{B,B}(t) = \begin{pmatrix} \vdots & & \vdots \\ \mathrm{Rep}_B(t(\vec{\beta}_1)) & \cdots & \mathrm{Rep}_B(t(\vec{\beta}_n)) \\ \vdots & & \vdots \end{pmatrix} = \begin{pmatrix} \lambda_1 & & 0 \\ \vdots & \ddots & \vdots \\ 0 & & \lambda_n \end{pmatrix}$$

Consider the representation of a member of this basis with respect to the basis $\mathrm{Rep}_B(\vec{\beta}_i)$. The product of the diagonal matrix and the representation vector

$$\mathrm{Rep}_B(t(\vec{\beta}_i)) = \begin{pmatrix} \lambda_1 & & 0 \\ \vdots & \ddots & \vdots \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ \lambda_i \\ \vdots \\ 0 \end{pmatrix}$$

has the stated action. QED

Eigenvalues and Eigenvectors

Eigenvalues and eigenvectors

Definition A transformation $t \colon V \to V$ has a scalar *eigenvalue* λ if there is a nonzero *eigenvector* $\vec{\zeta} \in V$ such that $t(\vec{\zeta}) = \lambda \cdot \vec{\zeta}$.

Eigenvalues and eigenvectors

Definition A transformation $t\colon V \to V$ has a scalar *eigenvalue* λ if there is a nonzero *eigenvector* $\vec{\zeta} \in V$ such that $t(\vec{\zeta}) = \lambda \cdot \vec{\zeta}$.

Definition A square matrix T has a scalar *eigenvalue* λ associated with the nonzero *eigenvector* $\vec{\zeta}$ if $T\vec{\zeta} = \lambda \cdot \vec{\zeta}$.

Note Any scalar multiple of an eigenvector is still an eigenvector.

Example The matrix

$$D = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$$

has an eigenvalue $\lambda_1 = 4$ and a second eigenvalue $\lambda_2 = 2$. The first is true because an associated eigenvector is $\vec{e_1}$

$$\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 4 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

and similarly for the second an associated eigenvector is e_2.

$$\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 2 \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

If we think of that matrix as representing a transformation of the plane, the transformation acts on those vectors in a particularly simple way, by rescaling.

But note that not every vector is simply rescaled.

$$\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \neq x \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Those that are rescaled, the eigenvectors, are special.

Computing eigenvalues and eigenvectors

Example We will find the eigenvalues and associated eigenvectors of this matrix.

$$T = \begin{pmatrix} 0 & 5 & 7 \\ -2 & 7 & 7 \\ -1 & 1 & 4 \end{pmatrix}$$

We want to find scalars x such that $T\vec{\zeta} = x\vec{\zeta}$ for some nonzero $\vec{\zeta}$. Bring the terms to the left side.

$$\begin{pmatrix} 0 & 5 & 7 \\ -2 & 7 & 7 \\ -1 & 1 & 4 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} - x \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and factor out the vector.

$$\begin{pmatrix} 0-x & 5 & 7 \\ -2 & 7-x & 7 \\ -1 & 1 & 4-x \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \qquad (*)$$

This homogeneous system has nonzero solutions if and only if the matrix is singular, that is, has a determinant of zero.

Some computation gives the determinant and its factors.

$$0 = \begin{vmatrix} 0-x & 5 & 7 \\ -2 & 7-x & 7 \\ -1 & 1 & 4-x \end{vmatrix}$$
$$= x^3 - 11x^2 + 38x - 40 = (x-5)(x-4)(x-2)$$

So the eigenvalues are $\lambda_1 = 5$, $\lambda_2 = 4$, and $\lambda_3 = 2$.

Some computation gives the determinant and its factors.

$$0 = \begin{vmatrix} 0-x & 5 & 7 \\ -2 & 7-x & 7 \\ -1 & 1 & 4-x \end{vmatrix}$$
$$= x^3 - 11x^2 + 38x - 40 = (x-5)(x-4)(x-2)$$

So the eigenvalues are $\lambda_1 = 5$, $\lambda_2 = 4$, and $\lambda_3 = 2$.

To find the eigenvectors associated with the eigenvalue of 5, specialize equation $(*)$ for $x = 5$.

$$\begin{pmatrix} -5 & 5 & 7 \\ -2 & 2 & 7 \\ -1 & 1 & -1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Gauss's Method gives this solution set; its nonzero elements are the eigenvectors.

$$V_5 = \{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} z_2 \mid z_2 \in \mathbb{C} \}$$

Similarly, to find the eigenvectors associated with the eigenvalue of 4, specialize equation $(*)$ for $x = 4$.

$$\begin{pmatrix} -4 & 5 & 7 \\ -2 & 3 & 7 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Gauss's Method gives this.

$$V_4 = \{ \begin{pmatrix} -7 \\ -7 \\ 1 \end{pmatrix} z_3 \mid z_3 \in \mathbb{C} \}$$

Similarly, to find the eigenvectors associated with the eigenvalue of 4, specialize equation $(*)$ for $x = 4$.

$$\begin{pmatrix} -4 & 5 & 7 \\ -2 & 3 & 7 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Gauss's Method gives this.

$$V_4 = \{ \begin{pmatrix} -7 \\ -7 \\ 1 \end{pmatrix} z_3 \mid z_3 \in \mathbb{C} \}$$

Specializing $(*)$ for $x = 2$

$$\begin{pmatrix} -2 & 5 & 7 \\ -2 & 5 & 7 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

gives this.

$$V_2 = \{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} z_3 \mid z_3 \in \mathbb{C} \}$$

Example If the matrix is upper diagonal or lower diagonal

$$T = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

then the polynomial is easy to factor.

$$0 = \begin{vmatrix} 2-x & 1 & 0 \\ 0 & 3-x & 1 \\ 0 & 0 & 2-x \end{vmatrix} = (3-x)(2-x)^2$$

Example If the matrix is upper diagonal or lower diagonal

$$T = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

then the polynomial is easy to factor.

$$0 = \begin{vmatrix} 2-x & 1 & 0 \\ 0 & 3-x & 1 \\ 0 & 0 & 2-x \end{vmatrix} = (3-x)(2-x)^2$$

These are the solutions for $\lambda_1 = 3$.

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad V_3 = \{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} z_2 \mid z_2 \in \mathbb{C} \}$$

These are for $\lambda_2 = 2$.

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad V_2 = \{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} z_1 \mid z_1 \in \mathbb{C} \}$$

Matrices that are similar have the same eigenvalues, but needn't have the same eigenvectors.

Example These two are similar

$$T = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 12 \end{pmatrix} \qquad S = \begin{pmatrix} 6 & -1 & -1 \\ 2 & 11 & -1 \\ -6 & -5 & 7 \end{pmatrix}$$

since $S = PTP^{-1}$ for this P.

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \end{pmatrix} \qquad P^{-1} = \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ -1/2 & 1/4 & 1/4 \\ -1/2 & -3/4 & 1/4 \end{pmatrix}$$

For the first matrix

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

is an eigenvector associated with the eigenvalue 4 but that does not hold for the second matrix.

Characteristic polynomial

Definition The *characteristic polynomial of a square matrix* T is the determinant $|T - xI|$, where x is a variable. The *characteristic equation* is $|T - xI| = 0$. The *characteristic polynomial of a transformation* t is the characteristic polynomial of any matrix representation $\mathrm{Rep}_{B,B}(t)$.

Characteristic polynomial

Definition The *characteristic polynomial of a square matrix* T is the determinant $|T - xI|$, where x is a variable. The *characteristic equation* is $|T - xI| = 0$. The *characteristic polynomial of a transformation* t is the characteristic polynomial of any matrix representation $\mathrm{Rep}_{B,B}(t)$.

Lemma A linear transformation on a nontrivial vector space has at least one eigenvalue.

Characteristic polynomial

Definition The *characteristic polynomial of a square matrix* T is the determinant $|T - xI|$, where x is a variable. The *characteristic equation* is $|T - xI| = 0$. The *characteristic polynomial of a transformation* t is the characteristic polynomial of any matrix representation $\text{Rep}_{B,B}(t)$.

Lemma A linear transformation on a nontrivial vector space has at least one eigenvalue.

Proof Any root of the characteristic polynomial is an eigenvalue. Over the complex numbers, any polynomial of degree one or greater has a root.

QED

Characteristic polynomial

Definition The *characteristic polynomial of a square matrix* T is the determinant $|T - xI|$, where x is a variable. The *characteristic equation* is $|T - xI| = 0$. The *characteristic polynomial of a transformation* t is the characteristic polynomial of any matrix representation $\mathrm{Rep}_{B,B}(t)$.

Lemma A linear transformation on a nontrivial vector space has at least one eigenvalue.

Proof Any root of the characteristic polynomial is an eigenvalue. Over the complex numbers, any polynomial of degree one or greater has a root.

QED

Remark This result is why we switched in this part from working with real number scalars to complex number scalars.

Eigenspace

Definition The *eigenspace of a transformation t associated with the eigenvalue λ* is $V_\lambda = \{\vec{\zeta} \mid t(\vec{\zeta}) = \lambda\vec{\zeta}\}$. The eigenspace of a matrix is analogous.

Example Recall that this matrix has three eigenvalues, 5, 4, and 2.

$$T = \begin{pmatrix} 0 & 5 & 7 \\ -2 & 7 & 7 \\ -1 & 1 & 4 \end{pmatrix}$$

Earlier, we found that these are the eigenspaces.

$$V_5 = \{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} c \mid c \in \mathbb{C}\} \quad V_4 = \{\begin{pmatrix} -7 \\ -7 \\ 1 \end{pmatrix} c \mid c \in \mathbb{C}\} \quad V_2 = \{\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} c \mid c \in \mathbb{C}\}$$

Lemma An eigenspace is a subspace. It is a nontrivial subspace.

Lemma An eigenspace is a subspace. It is a nontrivial subspace.

Proof (**Optional**) Notice first that V_λ is not empty; it contains the zero vector since $t(\vec{0})=\vec{0}$, which equals $\lambda\cdot\vec{0}$. To show that an eigenspace is a subspace, what remains is to check closure of this set under linear combinations. Take $\vec{\zeta}_1,\ldots,\vec{\zeta}_n\in V_\lambda$ and then

$$\begin{aligned}
t(c_1\vec{\zeta}_1+c_2\vec{\zeta}_2+\cdots+c_n\vec{\zeta}_n)&=c_1t(\vec{\zeta}_1)+\cdots+c_nt(\vec{\zeta}_n)\\
&=c_1\lambda\vec{\zeta}_1+\cdots+c_n\lambda\vec{\zeta}_n\\
&=\lambda(c_1\vec{\zeta}_1+\cdots+c_n\vec{\zeta}_n)
\end{aligned}$$

that the combination is also an element of V_λ.

The space V_λ contains more than just the zero vector because by definition λ is an eigenvalue only if $t(\vec{\zeta})=\lambda\vec{\zeta}$ has solutions for $\vec{\zeta}$ other than $\vec{0}$. QED

Theorem For any set of distinct eigenvalues of a map or matrix, a set of associated eigenvectors, one per eigenvalue, is linearly independent.

Theorem For any set of distinct eigenvalues of a map or matrix, a set of associated eigenvectors, one per eigenvalue, is linearly independent.

Proof (**Optional**) We will use induction on the number of eigenvalues. The base step is that there are zero eigenvalues. Then the set of associated vectors is empty and so is linearly independent.

Theorem For any set of distinct eigenvalues of a map or matrix, a set of associated eigenvectors, one per eigenvalue, is linearly independent.

Proof (**Optional**) We will use induction on the number of eigenvalues. The base step is that there are zero eigenvalues. Then the set of associated vectors is empty and so is linearly independent.

For the inductive step assume that the statement is true for any set of $k \geqslant 0$ distinct eigenvalues. Consider distinct eigenvalues $\lambda_1, \ldots, \lambda_{k+1}$ and let $\vec{v}_1, \ldots, \vec{v}_{k+1}$ be associated eigenvectors. Suppose that $\vec{0} = c_1\vec{v}_1 + \cdots + c_k\vec{v}_k + c_{k+1}\vec{v}_{k+1}$. Derive two equations from that, the first by multiplying by λ_{k+1} on both sides $\vec{0} = c_1\lambda_{k+1}\vec{v}_1 + \cdots + c_{k+1}\lambda_{k+1}\vec{v}_{k+1}$ and the second by applying the map to both sides $\vec{0} = c_1 t(\vec{v}_1) + \cdots + c_{k+1}t(\vec{v}_{k+1}) = c_1\lambda_1\vec{v}_1 + \cdots + c_{k+1}\lambda_{k+1}\vec{v}_{k+1}$ (applying the matrix gives the same result). Subtract the second from the first.

$$\vec{0} = c_1(\lambda_{k+1} - \lambda_1)\vec{v}_1 + \cdots + c_k(\lambda_{k+1} - \lambda_k)\vec{v}_k + c_{k+1}(\lambda_{k+1} - \lambda_{k+1})\vec{v}_{k+1}$$

The $\vec{v}_{k+1}$ term vanishes. Then the induction hypothesis gives that $c_1(\lambda_{k+1} - \lambda_1) = 0$, ..., $c_k(\lambda_{k+1} - \lambda_k) = 0$. The eigenvalues are distinct so the coefficients c_1, ..., c_k are all 0. With that we are left with the equation $\vec{0} = c_{k+1}\vec{v}_{k+1}$ so c_{k+1} is also 0. QED

Example This matrix from above has three eigenvalues, 5, 4, and 2.

$$T = \begin{pmatrix} 0 & 5 & 7 \\ -2 & 7 & 7 \\ -1 & 1 & 4 \end{pmatrix}$$

Picking a nonzero vector from each eigenspace we get this linearly independent set (which is a basis because it has three elements).

$$\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -14 \\ -14 \\ 2 \end{pmatrix}, \begin{pmatrix} -1/2 \\ 1/2 \\ -1/2 \end{pmatrix} \}$$

Example This matrix from above has three eigenvalues, 5, 4, and 2.

$$T = \begin{pmatrix} 0 & 5 & 7 \\ -2 & 7 & 7 \\ -1 & 1 & 4 \end{pmatrix}$$

Picking a nonzero vector from each eigenspace we get this linearly independent set (which is a basis because it has three elements).

$$\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -14 \\ -14 \\ 2 \end{pmatrix}, \begin{pmatrix} -1/2 \\ 1/2 \\ -1/2 \end{pmatrix} \}$$

Example This upper-triangular matrix has the eigenvalues 2 and 3

$$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Picking a vector from each of V_3 and V_2 gives this linearly independent set.

$$\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \}$$

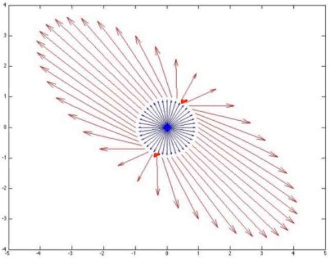
A criteria for diagonalizability

Corollary An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Geometry of eigenvectors

Geometrically, it turns out that multiplying the unit circle or unit sphere by a matrix A carves out an ellipse, or an ellipsoid. We can see eigenvectors visually by watching how multiplication by a matrix changes the unit vectors.

The following Figure illustrates this. The blue arrows represent some vectors the unit circle, all vectors for which $\|x\| = 1$. The red arrows represent the image of the blue arrows after multiplication by A, i.e. Ax. We can see how almost every vector changes direction when multiplied by A, except the eigenvector directions which are marked in black. Those are eigenvectors.



Power iteration for finding the largest eigenvector

Optional.

The power method is described by the following iterative operations:

$$v_{k+1} = \frac{Av_k}{\|Av_k\|} \tag{1}$$

And the converged solution will be largest eigenvector.